

Olivier LEBLANC

Doctor in applied maths to computational imaging
UCLouvain

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Professional Experience

- 2025 **Post-doctoral Researcher**, UCLouvain, Belgium
Computational imaging methods for gravitational wave detection.
- 2025 **Sabbatical year**, Martinique & South America
- 2020–2024 **Doctoral Researcher**, UCLouvain, Belgium
Compressive and neural-representation strategies for inverse problems in computational imaging (reducing the computational resources towards real-time endoscopy and scalable radio-interferometry, enhancing diffraction tomography by plugging AI-based signal representations into state-of-the-art physical models). Supervisor: Laurent Jacques.
- 2020–2024 **Teaching Assistant (TA)**, for two courses at UCLouvain
- *Image Processing and Computer Vision* (~50 students): designed, supervised, and graded various computer vision practical exams and exercise sessions in Python.
 - *EE master's project* (~50 students, 7 TAs + 4 professors): Designed and developed, for over 1.5 years, a prototype for a year-long autonomous audio-event classification project for environmental monitoring, co-led consultancy sessions, evaluations, and contests.
- 2018–2020 **Undergraduate TA**, coaching exercise sessions and grading homework
Signals and systems, electronics, electrical circuits, wave physics, and intro to quantum physics.
- Software Contributor to Open-Source Python toolboxes on Github: BASPlib, pyunlocbox, pyproximal.
- Scientific Publications and oral presentations in the leading journals and conferences of the field, both as an excellence autonomous researcher and as a team in international collaborations.
- Reviewing Regular contributions to peer-review (IEEE TCI, Audio Engineering Society, ISCS23, COSERA24...).

Education

- 2020–2024 **PhD in Engineering Sciences and Technology**, UCLouvain, Belgium
- 2018–2020 **Masters degree in electrical engineering**, UCLouvain, *magna cum laude*
Areas: signal and image processing, information theory, cryptography, machine learning, telecommunications, optics, aerospace dynamics, nanoelectronics, electromechanical converters, robotics.
- 2015–2018 **Bachelors degree in electromechanical engineering**, UCLouvain, *cum laude*

Awards

- Grant **Research Fellow**, *PhD funding for 4 years*, Fund for Scientific Research - FNRS, Belgium
- First prize **Best contribution award**, *for my contributions to multicore fiber lensless imaging*, BASP Frontiers conference 2023, Villars-sur-Ollon, Switzerland
- First prize **IEEE R8 Student Paper Contest 2021**, *for my experiments on walking droplets*, TELSIS 2021 conference, Niš, Serbia
- First prize **IEEE Best Master Thesis Award 2020**, *for my experiments on walking droplets*, UCLouvain

Professional Travels

- 2024 **Scientific stay**, *Two weeks of collaborative research on radio-interferometric imaging with the Biomedical and Astronomical Signal Processing (BASP) group led by Yves Wiaux*, Heriot-Watt University – Edinburgh, UK
- 2023 **Scientific stay**, *Two months of collaborative research on implicit neural representations with the Computational Imaging Group (CIG) led by U. Kamilov*, Washington University in St. Louis, USA
- 2022 **Scientific stay**, *One week of experimental measurements on an optical bench with a multicore fiber with the team of Hervé Rigneault*, Institut Fresnel – Aix-Marseille Université, France

2018 **Huawei**, *Two weeks to learn the ICT industry, research and suggest technologies to shape global sustainability efforts*, Beijing, Shenzhen, and Hong Kong, China

Skills

Languages **French** (mother tongue), **English** (C2), Dutch (B1), Spanish (B1)
Programming **Python**, **Matlab**, **C**, Java, HTML/CSS
Parallel **Pytorch**, **Matlab**, Cuda, Jax
Programming
Digital processing FPGA, Raspberry Pi, Arduino Uno, STM32 Nucleo board, Lime SDR, NXP FRDM-KL25Z
Data acquisition Accelerators, cameras, MOX (gas) sensors
Utilities LaTeX, git, Lt Spice, Labview, GNU Radio, photo and video editing, 3D CAD, Docker, common office software

Outside of work

Personality Curious and loves learning, sociable, good-humoured, conscientious
Hobbies Science, sport, reading, building and repairing stuff, music

Peer-reviewed publications

Journal papers

- [1] [Olivier Leblanc](#), Mathias Hofer, Siddharth Sivankutty, Hervé Rigneault and Laurent Jacques, "*Interferometric lensless imaging: rank-one projections of image frequencies with speckle illuminations*", IEEE Transactions on Computational Imaging, vol. 10, pp. 208–222, February 2024
- [2] Matteo Ravasi, Marcus Valtonen Örnå, Nick Luiken, [Olivier Leblanc](#), Eneko Uruñuela, "*PyProximal-scalable convex optimization in Python*", Journal of Open Source Software 9 (95), 6326, March 2024
- [3] Murielle Kirkove, Yuchen Zhao, [Olivier Leblanc](#), Laurent Jacques, Marc Georges, "*ADMM-inspired image reconstruction for terahertz off-axis digital holography*", JOSA A 41 (3), A1-A14, March 2024
- [4] [Olivier Leblanc](#), Yves Wiaux, Laurent Jacques, "*Compressive radio-interferometric sensing with random beamforming as rank-one signal covariance projections*", IEEE Transactions on Computational Imaging, vol. 11, pp. 1229–1242, July 2025
- [5] [Olivier Leblanc](#), Chung San Chu, Laurent Jacques, and Yves Wiaux, "*MROP: Modulated Rank-One Projections for compressive radio interferometric imaging*", Monthly Notices of the Royal Astronomical Society, vol. 543, n° 2, pp. 1727–1747, September 2025

Workshop abstracts

- [6] [Olivier Leblanc](#), Matthias Hofer, Siddharth Sivankutty, Hervé Rigneault, Laurent Jacques, "*An Interferometric view of Speckle Imaging*", Workshop on Low-Rank Models and Applications (LRMA), September 2022

Talks

- * "*Towards Interferometric Lensless Endoscopy: Rank-one Projections of Images Frequencies with Speckle Illuminations*", Institut Fresnel's group, Marseille (France), February 2022
- * "*Interferometric Lensless Imaging: Rank-one Projections of Image Frequencies with Speckle Illuminations*", Euler research building seminar, UCLouvain, February 2023
- * "*My background and tutorial on diffraction imaging*", Computational Imaging Group, Washington University in St. Louis (USA), May 2023
- * "*MultiCore Fiber Interferometric Imaging*", PoMM2024 : PhOtonique guidée Multimodale et Multi-canaux, Lille (France), May 2024